

BIMBINGAN TUGAS AKHIR #4

1 Marginal Likelihood

(Rasmussen & Williams, 2006, Bab 5.4.1, hlm. 112–113, Persamaan (5.7) & (5.8))

$$K_y = K + \sigma_n^2 I_N \quad (1)$$

$$\log p(\mathbf{y} \mid X, \boldsymbol{\theta}) = -\frac{1}{2} \mathbf{y}^T K_y^{-1} \mathbf{y} - \frac{1}{2} \log |K_y| - \frac{N}{2} \log(2\pi) \quad (2)$$

$$\frac{\partial \log p(\mathbf{y} \mid X, \boldsymbol{\theta})}{\partial \theta_j} = \frac{1}{2} \text{tr} \left(\left(\boldsymbol{\alpha} \boldsymbol{\alpha}^T - K_y^{-1} \right) \frac{\partial K_y}{\partial \theta_j} \right), \quad \boldsymbol{\alpha} = K_y^{-1} \mathbf{y} \quad (3)$$

2 Leave-One-Out Cross-Validation

(Rasmussen & Williams, 2006, Bab 5.3 & Bab 5.4.2, hlm. 111 & hlm. 116–118, Persamaan (5.6) & (5.11))

$$L_{\text{LOO}}(\boldsymbol{\theta}) = -\sum_{i=1}^N \log p(y_i \mid X, \mathbf{y}_{-i}, \boldsymbol{\theta}) = \frac{1}{2} \sum_{i=1}^N \left(-\log [K_y^{-1}]_{ii} + \frac{([K_y^{-1} \mathbf{y}]_i)^2}{[K_y^{-1}]_{ii}} + \log(2\pi) \right) \quad (4)$$

$$\frac{\partial L_{\text{LOO}}(\boldsymbol{\theta})}{\partial \theta_j} = \sum_{i=1}^n \left(\alpha_i [Z_j \boldsymbol{\alpha}]_i - \frac{1}{2} \left(1 + \frac{\alpha_i^2}{[K^{-1}]_{ii}} \right) [Z_j K^{-1}]_{ii} \right) / [K^{-1}]_{ii}. \quad (5)$$

Pustaka

- [1] Rasmussen, C. E., & Williams, C. K. (2006). *Gaussian Processes for Machine Learning*. MIT Press, Cambridge, MA.